

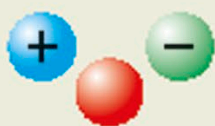
Physics

The branch of science that is concerned with matter and energy is called physics. This branch is central to all other sciences because it attempts to discover the basic laws that govern how the Universe works. Before the 20th century, physics concentrated mainly

on the areas of electricity, magnetism, force, motion, light, and waves. Today, it covers a wide range of fields, from thermodynamics to nuclear reactions.

What does it consist of?

Physics examines the behavior of matter and energy, which make up the entire Universe. It can be used to explain concepts in many areas of science such as meteorology, mechanics, and astronomy.



Particle physics

Particle physicists study the hundreds of different types of particles that make up atoms.



Mechanics

Mechanics is the study of the movement of objects and the forces that act upon them to set them in motion.



Wave theory

Wave theory tries to explain how natural phenomena, such as sound and light, are transmitted as waves.



Astronomy

This is the branch of physics that studies the planets, stars, and galaxies that make up the Universe.



Thermodynamics

This area of physics investigates the relationship between heat and other forms of energy.



Optics

Optics is the study of the behavior of beams of light as they reflect off or shine through different substances.



Electromagnetism

Electromagnetism studies the relationship between electric currents and magnetic fields.



Meteorology

Meteorologists study the weather, producing weather forecasts based on satellite and radar images.

SI units

Scientists use seven basic units of measurement, known as the SI base units, listed below. "SI" stands for "Système International." These units of measurement enable scientists in different countries to exchange the results of their experiments and calculations.

SI units		
Unit	Symbol	Quantity measured
Meter	m	A unit of length
Kilogram	kg	A unit of mass
Second	s	A unit of time
Ampere	A	A unit of electrical current
Kelvin	K	A unit of temperature
Candela	cd	A unit of brightness (luminous intensity)
Mole	mol	A unit of quantity of a substance (generally very small particles such as atoms and molecules)